Computational Astrophysics

→ Since the 70's Computer Simulations have become the third pillar of Science
↳ Together with Conventional Theory & Experiments

Direct Simulations vs. **Simulation of a model**

- $< \sim 10^6$ Particles interacting via relevant phisical laws
- Phisical Law is expressed by Field equation
  ↳ e.g. Poisson Equation for collisionless gravity system

- $> \sim 10^6$ Particles
  ↳ To much for even newest computers
- Discretize via usage of Superparticles or Meshpoints
- To be meaningful, the discretization has to be relying on robust mathematical Models

→ Distribution Function : $f(\vec{x}, \vec{v}, t)$

$f(\vec{x}, \vec{v}, t)$ describes how Phase Space $\vec{w}(\vec{x}, \vec{v})$ is filled by any given time

→ probability to have partical (or superpartical) with :

- position between $\vec{x}$ & $\vec{x} + d\vec{x}$
- velocity between $\vec{v}$ & $\vec{v} + d\vec{v}$
- at time t

$f(\vec{x}, \vec{v}, t)$ can describe Interaction in various fields :

→ Gravitational → Particles have Mass
→ Electromagnetical → Particles have Charge

→ Definition of Distribution Function:

$$F(\vec{r}, \vec{v}, t) = \sum_{i=1}^{N} \delta(\vec{r} - \vec{r}_i(t)) \cdot \delta(\vec{v} - \vec{v}_i(t))$$ → Exact distribution describing State

$$f(\vec{x}, \vec{v}, t) = \langle F(\vec{r}, \vec{v}, t) \rangle = \int F \, p \, d\vec{r}_1 \, d\vec{v}_1 \dots d\vec{r}_N \, d\vec{v}_n$$

↳ Avrg distribution
↳ = Exact distribution function weightet by probability density of states in phase space

→ **Uncorrelated Systems**

→ Evolution of a particle in Phase Space is independent of that of another particle at any time t

→ In a two-paticle System : $f_2(\vec{r}, \vec{v}, \vec{r}', \vec{v}', t) = f_1(\vec{r}, \vec{v}, t) \, f_1'(\vec{r}, \vec{v}, t)$

→ In a N-particle System : $f_N = f_1(\vec{r}_i, \vec{v}_i, t) \dots f_n(\vec{r}_n, \vec{v}_n, t)$

↳ Holds whenever direct interaction between forcefields are negligible yet particles still respond to mean forcefield
↳ e.g. Gravitational Systems

Colisionless Systems :

→ Gravitation between two particals is much smaller than theire kinetic energy → Two particles can come close but not change projectorys significantly

$$a_g = \frac{G m^2}{\frac{1}{2} m v^2} \sim (\sigma R)^{-1}$$

σ = Surface Density of for Example a Galaxy

R = Radius of Galaxy

$\epsilon_g = a_g \sigma^{\frac{1}{2}} \ll 1$ → Colisionless System

$\epsilon_p = a_p n^{\frac{1}{3}}$

→ Systems can change theyre collision propotys with time:

Colisionless System → t_{relax} → $\boxed{\text{Colisional System}}$ → Behaves like fluid

$$t_{relax} = \frac{N}{8 \ln N} t_{cross}$$

$$t_{cross} = \frac{R}{v} = \text{time to cross System}$$

$$v^2 = \frac{G N m}{R} \quad \rightarrow Nm = M = \text{total Mass}$$

$$= \frac{GM}{R}$$

→ Many Systems that can not be described as collision less can be described as pefect fluids

$$\phi(\vec{r}) = -G \sum_{j=1}^{N} \frac{m_j}{[(\vec{r}-\vec{r}_j)^2 + \varepsilon^2]^{\frac{1}{2}}}$$

→ Softening

↳ To avoid correlation between particles

↳ $\langle v^2 \rangle \gg \frac{Gm}{\varepsilon}$

→ Techniques to solve Colisionless Systems :

① Particle Mesh (PM) technique:

→ Construct auxilliary Mesh on which forces are computed by computing the potential

Steps: (I) Construction of a density field ρ from particle distribution

(II) Computation of the potential on the mesh by solving the Poisson - Equation with density field ρ
↳ Easier because in Fourier - Space

(III) Computation of the forcefield from the potential

(IV) Calculation of the forces at the original particle position